

Theorem 1. *For the quantum state discrimination problem, given the same set of mixed quantum states $\{\rho_1, \rho_2, \dots, \rho_M\}$ and the same prior probabilities $\{p_1, p_2, \dots, p_M\}$, the probability of successful discrimination $(P_{\text{succ}})^{\text{MED}} = (P_{\text{succ}})^{\text{MED}+}$. Also, $\Pi_{M+1} = 0$ when the equal sign holds.*

Proof.

Let POVM_M be the collection of POVMs that each POVM has M POVM elements, and here we allow a POVM element to be zero. Let $(P_{\text{succ}})^P$ denote the probability of successful discrimination of some POVM P or the POVM derived from some method P . We know that $(P_{\text{succ}})^{\text{MED}} \geq (P_{\text{succ}})^P, \forall P \in \text{POVM}_M$.

1. $(P_{\text{succ}})^{\text{MED}} \leq (P_{\text{succ}})^{\text{MED}+}$

Assume POVM_{MED} is an optimal solution for MED, and $\text{POVM}_{\text{MED}} \equiv \{\Pi_1, \Pi_2, \dots, \Pi_M\} \in \text{POVM}_M$. We can construct $P \equiv \{\Pi'_1, \Pi'_2, \dots, \Pi'_{M+1}\} \in \text{POVM}_{M+1}$ by

$$\Pi'_1 = \Pi_1, \Pi'_2 = \Pi_2, \dots, \Pi'_M = \Pi_M, \Pi'_{M+1} = 0$$

Then,

$$(P_{\text{succ}})^{\text{MED}+} \geq (P_{\text{succ}})^P = \sum_{i=1}^M p_i \text{tr}(\rho_i \Pi'_i) \quad (1)$$

$$= \sum_{i=1}^M p_i \text{tr}(\rho_i \Pi_i) \quad (2)$$

$$= (P_{\text{succ}})^{\text{MED}} \quad (3)$$

2. $(P_{\text{succ}})^{\text{MED}} \geq (P_{\text{succ}})^{\text{MED}+}$

Assume $\text{POVM}_{\text{MED}+}$ is an optimal solution for MED+, and $\text{POVM}_{\text{MED}+} \equiv \{\Pi_1, \Pi_2, \dots, \Pi_{M+1}\} \in \text{POVM}_{M+1}$. We can construct $P \equiv \{\Pi'_1, \Pi'_2, \dots, \Pi'_M\} \in \text{POVM}_M$ by

$$\Pi'_1 = \Pi_1, \Pi'_2 = \Pi_2, \dots, \Pi'_M = \Pi_M + \Pi_{M+1}$$

Then,

$$(P_{\text{succ}})^{\text{MED}} \geq (P_{\text{succ}})^P = \sum_{i=1}^M p_i \text{tr}(\rho_i \Pi'_i) \quad (4)$$

$$= \sum_{i=1}^M p_i \text{tr}(\rho_i \Pi_i) + p_M \text{tr}(\rho_i \Pi_{M+1}) \quad (5)$$

$$= (P_{\text{succ}})^{\text{MED}+} + p_M \text{tr}(\rho_i \Pi_{M+1}) \quad (6)$$

$$\geq (P_{\text{succ}})^{\text{MED}+} \quad (7)$$

3. By 1. and 2., we can see that $(P_{\text{succ}})^{\text{MED}} = (P_{\text{succ}})^{\text{MED}+}$.

4. To make the equal sign in Eq (7) hold, $\Pi_{M+1} = 0$.

□

From Theorem 1, we can easily prove the following corollary.

Corollary 1. *For the quantum state discrimination problem, given the same set of non-orthogonal, linearly independent pure quantum states $\{|\psi_1\rangle, |\psi_2\rangle, \dots, |\psi_M\rangle\}$ and the same prior probabilities $\{p_1, p_2, \dots, p_M\}$, the probability of successful discrimination $(P_{\text{succ}})^{\text{MED}} \geq (P_{\text{succ}})^{\text{OptUQSD}}$.*

Proof. If we see the MED+ problem and the Optimal UQSD problem as an convex optimization problem, then they have the same variables and the same objective function (P_{succ}) , with the Optimal UQSD problem posing extra constraints on the error terms ($P_{\text{err}} = 0$). This means that the optimal solution of OptUQSD is always a feasible point of MED+, so

$$(P_{\text{succ}})^{\text{MED}+} \geq (P_{\text{succ}})^{\text{OptUQSD}}. \quad (8)$$

Then

$$(P_{\text{succ}})^{\text{MED}} = (P_{\text{succ}})^{\text{MED}+} \geq (P_{\text{succ}})^{\text{OptUQSD}}. \quad (9)$$

□